DICTO: Collected Geometric Works

Papers, findings and open apps on polyhedra, space-filling and icosahedral order, by DICTO

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Version 17 · 20 October 2026 · DICTO DOI: 10.5281/zenodo.23345699 (all versions) ·
Licence CC BY-SA 4.0 · The apps: kaleidohedra.dictospheres.com, rhombiverse.dictospheres.com, polyhedraverse.dictospheres.com.
All works, findings and names are DICTO's. Computation and checking with Claude (Anthropic).

About. DICTO is a geometric sculptor in Osaka whose work grows out of building with recycled PET and from three open apps on one shared geometry, Kaleidohedra, Rhombiverse and Polyhedraverse. This record gathers DICTO's published geometry under one DOI. Each work below has its own DICTO DOI, always pointing to its latest version; the numbers are those of DICTO's findings on dictospheres.com. Version 17 adds #40, The DICTO OG Family on Every Stacking, and #39, The DICTO OG Family: Four Steps Up and Four Down, and lists Version 3 of #38 (new figures, and the prior art for its rule) and Version 4 of #37a (it is the DICTO Jewel's packing, not the solid, that does not twin); Version 16 listed the Version 3 updates of #37a (the DICTO Octahedrally Truncated Triacontahedron and the DICTO Fill sets) and #31a (the OCD Stella's crystals); Version 15 added #37a, The DICTO Ogdoecontahedron on the Gyrated Octet; Version 14 added #38a, The DICTO Kepler–Poinsot–UP Cycle, and recorded that from 20 October 2026 every paper listed here is released under CC BY-SA 4.0 (Attribution-ShareAlike): each has a new version with that licence, and earlier versions keep CC BY 4.0. Version 13 added #38, Universal PiTet: The DICTO PiTet Proposition; Version 12 added #37, The DICTO Ogdoecontahedron on the Octet Truss; Version 11 added #36, Golden Bodhi Pyrochlore; Version 10 corrected the paper count and named the seven findings published with the app; Version 9 added #34a, The DICTO Hecatogdoecontahedron, and #31a, The DICTO OCD Stella (the third and fourth short introductions, one per new solid); Version 8 added #35b, The DICTO Palintriacontahedron (the second); Version 7 added #35a, The DICTO Metaxyhexecontahedron (the first); Version 6 added #35, Vajrayana; Version 5 added #34, Progenitors; Version 4 lists the Version 2 updates of #28, #31, #32 and #33; Version 3 added Double Vision (#33); Version 2 added the twelve papers published or given their own DOI since Version 1 (14 October 2026).

Papers (32), newest first

#	Work	About	DOI
#40	**The DICTO OG Family on Every Stacking** (2026)	One exact kit builds the octet, the gyrated octet, the classical polytypes and a Fibonacci crystal that never repeats. Two short rules say why: the twin rule (the mirror twin is the solid turned 60°, the spinel law) and the fit rule (a solid that stays in its layer fits every stacking); the DBT fits every stacking, and the golden envelope fails at twin planes by exactly the tips the DBT removes.	DICTO DOI: 10.5281/zenodo.23490114

#39	**The DICTO OG Family: Four Steps Up and Four Down** (2026)	Universal PiTet's step run from the Ogdoecontahedron, outward and in reverse: eight new icosahedral solids, the DICTO KRE, PHE, TPP and DPHP up and the DICTO TPH, TQH, TQP and TQHD down, all exact; near the OG the shells nest and fill exactly, then both directions break the same way.	DICTO DOI: 10.5281/zenodo.23488944
#38a	**The DICTO Kepler–Poinsot–UP Cycle** (2026)	The tall companion to Universal PiTet: the great and small stellated dodecahedra in an endless φ^3 cycle, every shell filled exactly by one pattern, in golden reciprocity.	DICTO DOI: 10.5281/zenodo.23464021
#38	**Universal PiTet: The DICTO PiTet Proposition** (2026)	From the Cleo star outward in spherical shells: an endless, self-polar cycle of four solids, icosidodecahedron, rhombic triacontahedron, DICTO GTI and DICTO FPD, closing exactly $10/\varphi^4$ larger each turn, with exact shell pieces, Monkey Staff B, five-sided Vajras and hexagonal Cleos. Version 3: new figures, and the rule's prior art (the three-dimensional antipedal construction).	DICTO DOI: 10.5281/zenodo.23463305
#37a	**The DICTO Ogdoecontahedron on the Gyrated Octet** (2026)	The companion to #37: in the gyrated octet the OGs twin across every layer (the OG turned 60°, the spinel law); one exact cell kit for every stacking, with twin caps, bipyramids and a mirror middle piece; the OCD Stella twins too, and the DICTO Belted Triacontahedron fills the octahedra pole to pole; the DICTO Octahedrally Truncated Triacontahedron fits every stacking, and each solid with one fill piece fills space. Version 4: it is the DICTO Jewel's packing, not the solid, that does not twin.	DICTO DOI: 10.5281/zenodo.23484012
#37	**The DICTO Ogdoecontahedron on the Octet Truss** (2026)	Exact space-filling kits with the OG in the octahedra of the octet truss: golden-section keys; an exact Penrose link between facing five-fold stars, with three Penrose cores; golden envelopes filling exactly $10/\varphi^6$ of space, with a two-shape kit and a tetartoid core; and one-piece glue with exact OG imprints, keyed with the OG's own kites so that twelve lock round one OG.	DICTO DOI: 10.5281/zenodo.23457016
#36	**Golden Bodhi Pyrochlore** (2026)	The gap of the Ogdoecontahedron's bcc packing, exact: 72 cone pieces, limbs crossing in golden rhombi, the canonical golden cut that shatters, the GBPC that every golden fill needs, and no face-to-face tiling of what remains by golden rhombohedra; DICTO's Bull-horns OG adaptor fills the whole gap with one piece, three per copy.	DICTO DOI: 10.5281/zenodo.23451210

#35b	**The DICTO Palintriacontahedron** (2026)	A short introduction: the icosidodecahedron's central pyramids flipped out at full height; the hull of the 32 tips, 60 triangles, each half a rhombic-dodecahedron rhombus: the member of the dodecahedron–icosahedron hull family at edge ratio $3/\sqrt{5}$, exact. It interlocks with the next icosahedron, bridging one turn of the golden cycle to the next.	DICTO DOI: 10.5281/zenodo.23430866
#35a	**The DICTO Metaxyhexecontahedron** (2026)	A short introduction: the hull of an icosahedron and a dodecahedron at edge ratio $9/\varphi^4$, in closed form, exact; the solid that closes the golden cycle's last gap, touching the rhombic triacontahedron and the next icosahedron.	DICTO DOI: 10.5281/zenodo.23430864
#35	**Vajrayana** (2026)	A companion to Progenitors: the dodecahedron returns inside the golden cycle as the hull of DICTO's Cleo star, filled exactly by 82 pieces of four kinds; the last loose step is closed by a new solid that touches both sides, the DICTO Metaxyhexecontahedron, each of its two layers one kind of piece; the cycle grows to twelve stations.	DICTO DOI: 10.5281/zenodo.23429911
#34a	**The DICTO Hecatogdoecontahedron** (2026)	A short introduction: the DICTO Hecatodihedron and its polar, the DICTO Ogdoecontahedron, interlocked in the truncated icosahedron's midsphere as one self-dual compound, 182 faces and 182 corners; core and hull are the classical rectified and joined truncated icosahedron; the pairing and the name are what is new.	DICTO DOI: 10.5281/zenodo.23430868
#34	**Progenitors** (2026)	The DICTO Hecatodihedron and Ogdoecontahedron as a twin pair: on their own they never close golden; set into the Spheres of Dharma in the dodecahedron's place they close it again on φ^9, and the cycle holds the core and hull of two dual compounds.	DICTO DOI: 10.5281/zenodo.23413184
#33	**Double Vision – Both Cycles in One Space** (2026)	The Spheres of Dharma and their inside-out reversal in one space: one sphere of reciprocation toggles which shells the two cycles share, and no sphere gives both. Cut against each other they make one solid of exact convex pieces, 536 of 12 kinds per turn in one setting and 502 of 11 in the other.	DICTO DOI: 10.5281/zenodo.23407107
#32	**Inverted Shells: The Spheres of Dharma Turned Inside Out** (2026)	Polar reciprocation turns the Spheres of Dharma inside out into a new exact golden cycle, every layer filled by one kind of piece. The great stellated dodecahedron locks into its outer gap at exactly one size, a dual pair fits its inner gap in reverse order, and the cycle repeats inward without end.	DICTO DOI: 10.5281/zenodo.23405029

#31a	**The DICTO OCD Stella** (2026)	A short introduction: the icosahedral hull of the golden adapter pieces round one Ogdoecontahedron in its body-centred cubic packing, cut along the 8 joints: 48 corners, 50 faces; copies fill exactly $151/2 - 267\sqrt{5}/8 \approx 87.12\%$ of space, and twelve round a small one fill an outer one exactly at φ^2; with regular octahedra on the second network its crystal fills $1423/32 - 3113\sqrt{5}/160 \approx 96.33\%$.	DICTO DOI: 10.5281/zenodo.23430870
#31	**The DICTO Ogdoecontahedron** (2026)	The Spheres of Dharma turned inside out: the polar dual of the Hecatodihedron, a convex solid of 20 regular hexagons and 60 kites, exact in every coordinate. Joined on its hexagons it forms pyrochlore networks and a body-centred cubic packing filling exactly 77.43% of space.	DICTO DOI: 10.5281/zenodo.23404688
#30	**A Fibonacci Spine** (2026)	Five periodic packings of the DICTO Ogdoecontahedron meet round a five-fold axis; with the central copy, the copies removed along the axis are exactly the canonical Fibonacci chain, checked exactly, until minimal removal thins it at height 12.8.	DICTO DOI: 10.5281/zenodo.23396361
#29	**Theaetetus** (2026)	An icosahedron held exactly by its Cleo-Icos, grown to the DICTO Hecatodihedron in two exact stories: 215 pieces in closed layers, and 102 clean pieces that cut the icosahedron and dodecahedron with the same twenty cones.	DICTO DOI: 10.5281/zenodo.23378070
#28	**The DICTO Truncated-Icosahedral Hecatodihedron** (2026)	A convex solid of 102 faces from the Spheres of Dharma: the truncated icosahedron with a low pyramid on each hexagon, at the height where neighbouring triangles flatten into rhombi; exactly, the joined truncated icosahedron with its five-fold corners cut flat. Version 2 adds the DICTO Hecatogdoecontahedron.	DICTO DOI: 10.5281/zenodo.23360857
#27	**Capturing the In-betweens** (2026)	The in-between stages of the Chain of Karma, Spheres of Dharma and Bodhi Seed as solids: the great stellated dodecahedron, the excavated dodecahedron, the rhombic hexecontahedron, and a further convex solid, the DICTO Truncated-Icosahedral Hecatodihedron.	DICTO DOI: 10.5281/zenodo.23345502
#26	**DICTO ATOM-13×13** (2026)	Thirteen clusters of twelve around one, six golden and seven periodic, side by side, with 24 buildable kits.	DICTO DOI: 10.5281/zenodo.23333183
#25	**DICTO Spheroid-13 RD** (2026)	A true rhombic dodecahedron wrapped by twelve keyed pieces: a nearly round block of thirteen that fills space by translation alone.	DICTO DOI: 10.5281/zenodo.23332821
#24	**The Bodhi Seed** (2026)	Twelve rhombic triacontahedra and one more inside a triacontahedron φ^2 times larger; the gap fills exactly with three convex piece types, in two ways.	DICTO DOI: 10.5281/zenodo.23311353

#23	**Spheres of Dharma** (2026)	Six icosahedral shells, each filled exactly, repeating every φ^3: icosahedron, icosidodecahedron, dodecahedron, fullerene cage, rhombic triacontahedron.	DICTO DOI: 10.5281/zenodo.23310833
#22	**The Chain of Karma** (2026)	The regular icosahedron and dodecahedron filled exactly as shells of each other, alternating without end (built and checked to 20,497 pieces). The centres of the Lotus seed's 13 icosahedra match the 600-cell's central vertex and its 12 neighbours, scaled by φ^2; and its surface has the C60 fullerene pattern. Only the icosahedron is ever subdivided; the gap pieces occur at every scale.	DICTO DOI: 10.5281/zenodo.23310541
#20	**The FUJI Family** (2026)	FUJI as one member of a family of pyramids on flat-ended triangles, indexed by the antiprisms. With the golden breathing pyrochlore, the wheel bound, and Cleo-Icos & Antonius, a new space-filling pair from the golden rhombohedron.	DICTO DOI: 10.5281/zenodo.23357896
#19	**AXE, CLEO and FUJI** (2026)	The three pieces that close DICTO's thirteen-icosahedron cluster into one icosahedron of edge φ^2. AXE and CLEO are Mosseri–Sadoc tiles; FUJI is new. With printable nets.	DICTO DOI: 10.5281/zenodo.23351291
#18	**The DICTO Hexa** (2026)	A cube of 8 DICTO Hexa Jewels, its Hexa-Key and their 5 : 3 checkerboard.	DICTO DOI: 10.5281/zenodo.23376320
#17	**Kagome Hulls** (2026)	Six corner-sharing networks of solid clusters, one rule and its icosahedral exception.	DICTO DOI: 10.5281/zenodo.23376317
#14	**The DICTO-Star** (2026)	An icosidodecahedron, 12 dodecahedra and 20 tridiminished icosahedra closing exactly.	DICTO DOI: 10.5281/zenodo.23284213
#13	**The DICTO Dodeca-13** (2026)	The 13-dodecahedron cluster made solid, with two exact gap pieces.	DICTO DOI: 10.5281/zenodo.23284281
#8	**The Euclid–Kepler–Pacioli Cell** (2026)	All five Platonic solids nest on 13 points in one periodic cell.	DICTO DOI: 10.5281/zenodo.23284106

Findings and apps

Work	About	DOI
Kaleidohedra by DICTO (2026)	DICTO's findings (DISCOVERIES #1 onwards) with the open apps' code, in versioned releases. Findings published there without a separate paper: #1 DICTO skewed rhombic dodecahedron; #5 Regular-hexagon elongated dodecahedron (found independently; known, Grünbaum 2010); #10 DICTO Jewel (EKP windows); #11 The stretched dodecahedron; #12 Star Chain Reaction; #15 DICTO Jewel clusters; #16 Sunstar clusters	DICTO DOI: 10.5281/zenodo.23173809

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